



1. **DESCRIPTION:** Teams will complete tasks related to the properties and processes of Earth's fresh waters.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class II

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
- b. Each team may bring two Class II calculators.
- c. Each team may bring a binder of any size containing information in any form and from any source. Sheet protectors, lamination, tabs, and labels are permitted. If the event features a rotation through a series of laboratory stations where the participants interact with samples, specimens, or displays, no material may be removed from the binder throughout the event.

3. **THE COMPETITION:** Participants will be assessed through an exam and/or timed stations on the following topics:

- a. Hydrologic cycle, water budgets, and the critical zone
 - i. Components and associated fluxes of the hydrologic cycle (e.g., precipitation, evaporation, transpiration, evapotranspiration, infiltration, runoff, condensation)
 - ii. Storage reservoirs, residence times, and water budgets
- b. Climatic controls on freshwater distribution
 - i. Precipitation and atmospheric inputs to freshwater systems (e.g., rainfall, snowfall, seasonality, intensity, spatial patterns)
 - ii. Topographic and geographic controls on precipitation and climate (e.g., orographic lift and the rain shadow effect, continentality, lake effect, elevation)
 - iii. The arid-semiarid-humid spectrum and its hydrologic and geomorphic expression (e.g., perennial vs. ephemeral streams, playas, evaporite deposits)
- c. Drainage basins and stream networks
 - i. Watersheds, drainage divides, and drainage basin characteristics
 - ii. Stream order and networks
 - iii. Drainage patterns, density, and texture
- d. Fluvial sediment processes and channel form
 - i. Channel types and planform (e.g., braided, meandering, straight, anastomosing) and calculation of sinuosity
 - ii. Sediment properties (e.g., grain size, shape, sorting, rounding)
 - iii. Sediment transport and deposition (e.g., bed load, suspended load, dissolved load) and the controls on sediment supply, competence, and capacity
 - iv. Fluvial bedforms (e.g., ripples, dunes, antidunes, bars)
- e. Fluvial geomorphology — landforms and valley processes
 - i. Longitudinal profile, gradient, and base level
 - ii. Dynamic equilibrium and graded streams
 - iii. Erosional features (e.g., knickpoints, waterfalls, v-shaped valleys)
 - iv. Erosional processes (e.g. stream capture, downcutting, abrasion, meandering, hydraulic action)
 - v. Channel and floodplain landforms (e.g., point bars, cut banks, oxbow lakes, floodplains, natural levees, deltas, alluvial fans)
- f. Streamflow and floods
 - i. Flow regimes (e.g., perennial, intermittent, ephemeral)
 - ii. Discharge, velocity, and load (water and sediment)
 - iii. Mathematical relationships describing open-channel flow
 - iv. Floods, flood frequency, and recurrence intervals
- g. Groundwater and karst
 - i. Subsurface zones (e.g., zone of aeration, zone of saturation, water table)
 - ii. Aquifers and their properties (e.g., confined, unconfined, and artesian aquifers; porosity, permeability)
 - iii. Hydraulic gradient and groundwater flow
 - iv. Recharge, discharge, and capillarity
 - v. Interactions between surface water and groundwater, including saltwater intrusion
 - vi. Karst processes, hydrology, landforms, and features (e.g., sinkholes, solution valleys, caves, disappearing streams, springs)



- h. Lakes and wetlands
 - i. Lake formation and types (e.g., tectonic/rifting, volcanic, glacial, fluvial damming)
 - ii. Lake water budgets — inflow and outflow
 - iii. Physical and chemical properties and stratification (e.g., seasonal turnover)
 - iv. Shoreline processes (e.g., waves, shoreline features)
 - v. Wetland types, formation, and hydrology (e.g., bogs, marshes, swamps, fens)
 - i. Geologic and paleohydrologic record
 - i. Sedimentary structures as indicators of water and flow conditions (e.g., ripple marks, raindrop imprints, mud cracks, cross-bedding, graded bedding)
 - ii. Paleochannels and preserved fluvial and lacustrine deposits
 - j. Human impacts and water quality
 - i. Effects of land-use change on freshwater systems (e.g., altered runoff, sedimentation, ecological change)
 - ii. Engineering structures, water extraction, and their effects (e.g., dams, levees, channelization, diversion, downcutting; wells and groundwater pumping, with effects such as aquifer depletion and land subsidence)
 - iii. Water pollution (e.g., types, sources, transport)
 - k. Measuring and monitoring freshwater
 - i. Interpreting freshwater features on topographic maps
 - ii. Mapping the water table from well data (e.g., contour maps, flow lines, flow direction)
 - iii. Stream gauging, discharge measurement, and hydrographs (e.g., stage–discharge relationships, baseflow vs. stormflow, peak discharge, lag time)
 - iv. Monitoring networks and techniques (e.g., groundwater observation wells, tracer studies, remote sensing)
4. **SCORING:**
- a. High Score wins.
 - b. Points will be awarded for the quality and accuracy of answers, the quality of supporting reasoning, and the use of proper scientific methods in responses.
 - c. Preselected questions will be used as tiebreakers.

In partnership with the National Oceanic and Atmospheric Administration (NOAA) and the North American Association for Environmental Education (NAAEE)